3.5 Smooth Manifolds — Solutions

Solutions to problems.md. Theorems are quoted from Lee (2nd ed.) or notes.md by name.

Warm-up

1. Stereographic atlas

For $u \in \mathbb{R}^n$,

$$\sigma_N^{-1}(u) = \left(\frac{2u}{|u|^2+1}, \frac{|u|^2-1}{|u|^2+1} \right).$$

Check: this point has norm $\frac{4|u|^2+(|u|^2-1)^2}{(|u|^2+1)^2} = 1$, and with $x^{n+1} = \frac{|u|^2-1}{|u|^2+1}$ we get $1 - x^{n+1} = \frac{2}{|u|^2+1}$, so σ_N of it is $\frac{2u}{|u|^2+1} \cdot \frac{|u|^2+1}{2} = u$. Conversely $\sigma_N^{-1} \circ \sigma_N = \mathrm{id}$ by the same computation (or because σ_N is injective: the line through N and x meets the sphere in only one other point). Both maps are continuous, so σ_N is a homeomorphism onto $\mathbb{R}^n$; similarly for σ_S.

The overlap is $S^n \setminus \{N, S\}$, and σ_N maps it onto $\mathbb{R}^n \setminus \{0\}$. For $u \neq 0$, $1 + x^{n+1} = \frac{2|u|^2}{|u|^2+1}$, so

$$\sigma_S \circ \sigma_N^{-1}(u) = \frac{2u}{|u|^2+1} \cdot \frac{|u|^2+1}{2|u|^2} = \frac{u}{|u|^2}.$$

This is smooth on $\mathbb{R}^n \setminus 0$ and is its own inverse, so it is a diffeomorphism. The two charts cover S^n, so they form a smooth atlas. □

2. Hyperboloids and the cone

(a) $df_{(x,y,z)} = 2x\,dx + 2y\,dy - 2z\,dz$ vanishes only at the origin, and $f(0) = 0$. So every $c \neq 0$ is a regular value and $f^{-1}(c)$ is an embedded 2-dimensional submanifold (a one-sheeted hyperboloid for $c > 0$, two-sheeted for $c < 0$). $c = 0$ is a critical value.

(b) Let $C = f^{-1}(0)$. Suppose 0 has an open neighborhood $W \subseteq C$ with a homeomorphism $h : W \to \mathbb{R}^2$. Choose an open disc D around $h(0)$; then $V = h^{-1}(D)$ is an open neighborhood of 0 in C and $V \setminus \{0\} \cong D \setminus \{h(0)\}$ is connected. Since V is open in C and contains 0, it contains $(t, 0, t)$ and $(t, 0, -t)$ for all small $t > 0$. But z is continuous and nonzero on $C \setminus \{0\}$ (if $z = 0$ then $x = y = 0$), so it has constant sign on the connected set $V \setminus \{0\}$. Contradiction. So C is not a topological manifold near 0, and in particular not an embedded submanifold. □

3. A 2-form on $\mathbb{R}^3$

(a) $d\omega = dx \wedge dy \wedge dz + dy \wedge dz \wedge dx + dz \wedge dx \wedge dy = 3\,dx \wedge dy \wedge dz$, since cyclic permutations of three letters are even.

(b) Using $dx^i \wedge dx^j(u, v) = u^i v^j - u^j v^i$:

$$\omega_p(u, v) = x(u^2v^3 - u^3v^2) + y(u^3v^1 - u^1v^3) + z(u^1v^2 - u^2v^1) = p \cdot (u \times v).$$

(c) $F^*\omega = h\,d\varphi \wedge d\theta$ with $h = (F^*\omega)(\partial_\varphi, \partial_\theta) = \omega_F(F_\varphi, F_\theta) = F \cdot (F_\varphi \times F_\theta)$. With $F_\varphi = (\cos\varphi\cos\theta, \cos\varphi\sin\theta, -\sin\varphi)$ and $F_\theta = (-\sin\varphi\sin\theta, \sin\varphi\cos\theta, 0)$:

$$F_\varphi \times F_\theta = (\sin^2\varphi\cos\theta,\ \sin^2\varphi\sin\theta,\ \sin\varphi\cos\varphi),$$

and $F \cdot (F_\varphi \times F_\theta) = \sin^3\varphi\cos^2\theta + \sin^3\varphi\sin^2\theta + \sin\varphi\cos^2\varphi = \sin\varphi$. So

$$F^*\omega = \sin\varphi\, d\varphi \wedge d\theta. \qquad \square$$

4. Brackets

Use $[X,Y]^j = X^i\partial_i Y^j - Y^i\partial_i X^j$. Write $X = (-y, x)$, $Y = (1, 0)$, $Z = (x, y)$.

- $[X, Y]$: Y is constant so the first term vanishes; $Y^i\partial_i X = \partial_x X = (0, 1)$. So $[X, Y] = -\partial_y$.
- $[X, Z]$: $X^i\partial_i Z = X$ (since $DZ = I$), and $Z^i\partial_i X = DX \cdot Z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = X$. So $[X, Z] = 0$.

Interpretation. X generates rotations θ_t and Z generates dilations $\psi_s(p) = e^s p$. Rotations commute with scalings, consistent with $[X, Z] = 0$ (Problem 21). The translations generated by Y do not commute with rotations, and $[X, Y] \neq 0$ records this. $\square$

5. The angle form

Write $\omega = P\,dy + Q\,dx$ with $P = x/r^2$, $Q = -y/r^2$, $r^2 = x^2 + y^2$. Then $d\omega = (\partial_x P - \partial_y Q)\,dx \wedge dy$ and

$$\partial_x P = \frac{r^2 - 2x^2}{r^4} = \frac{y^2 - x^2}{r^4}, \qquad \partial_y Q = -\frac{r^2 - 2y^2}{r^4} = \frac{y^2 - x^2}{r^4},$$

so $d\omega = 0$.

With $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$: $\gamma^*\omega = (\cos t \cdot \cos t - \sin t \cdot (-\sin t))\,dt = dt$, so $\int_\gamma \omega = 2\pi$. If $\omega = df$, then $\int_\gamma df = f(\gamma(2\pi)) - f(\gamma(0)) = 0$. So ω is not exact.

On $\{x > 0\}$, $\vartheta = \arctan(y/x)$ is smooth and $d\vartheta = \dfrac{1}{1 + y^2/x^2}\left(\dfrac{dy}{x} - \dfrac{y\,dx}{x^2}\right) = \dfrac{x\,dy - y\,dx}{x^2 + y^2} = \omega$. $\square$

6. $O(n)$

Let $\mathrm{Sym}(n)$ be the symmetric matrices, a vector space of dimension $n(n+1)/2$, and $\Phi : GL(n) \to \mathrm{Sym}(n)$, $\Phi(A) = A^TA$. Then $O(n) = \Phi^{-1}(I)$ and

$$d\Phi_A(B) = \frac{d}{dt}\Big|_0 (A + tB)^T(A + tB) = B^TA + A^TB.$$

For $A \in O(n)$ and $C \in \mathrm{Sym}(n)$, put $B = \frac{1}{2}AC$. Then $d\Phi_A(B) = \frac{1}{2}C^TA^TA + \frac{1}{2}A^TAC = \frac{1}{2}C + \frac{1}{2}C = C$. So I is a regular value, and $O(n)$ is an embedded submanifold of dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$, with

$$T_I O(n) = \ker d\Phi_I = \{B : B^T + B = 0\},$$

the skew-symmetric matrices (dimension $n(n-1)/2$, as it must be). $\square$

Core

7. Orientability of $\mathbb{RP}^n$

Let $\mu = dx^0 \wedge \cdots \wedge dx^n$ on $\mathbb{R}^{n+1}$, $N = x^i \partial_i$ the position field, and $\Omega = \iota^*(i_N \mu)$, where $\iota : S^n \hookrightarrow \mathbb{R}^{n+1}$. Ω is a volume form on S^n: for a basis $v_1, \dots, v_n$ of $T_x S^n = x^\perp$, $\Omega(v_1, \dots, v_n) = \mu(x, v_1, \dots, v_n) \neq 0$.

Let $a(x) = -x$. As a linear map on $\mathbb{R}^{n+1}$, $a^*\mu = \det(-I)\mu = (-1)^{n+1}\mu$, and $da_x(N_x) = -x = N_{-x}$, so N is a-related to itself. Hence $a^*(i_N \mu) = i_N(a^*\mu)$, and restricting to the sphere,

$$a^*\Omega = (-1)^{n+1}\Omega.$$

Let $q : S^n \to \mathbb{RP}^n$ be the quotient, a smooth 2-sheeted covering and local diffeomorphism with $q \circ a = q$.

n odd. Then $a^*\Omega = \Omega$. Define ν on $\mathbb{RP}^n$ by $\nu_{q(x)} = \left((dq_x)^{-1}\right)^* \Omega_x$ on a neighborhood where q has a local inverse. The two choices $x, -x$ give the same form because $a^*\Omega = \Omega$ and $q \circ a = q$. So ν is a well-defined smooth nowhere-vanishing n-form and $\mathbb{RP}^n$ is orientable.

n even. Suppose ν is a volume form on $\mathbb{RP}^n$. Then $q^*\nu$ is nowhere vanishing on S^n, so $q^*\nu = f\Omega$ with f smooth and never zero; S^n is connected, so f has constant sign. From $q \circ a = q$, $a^*q^*\nu = q^*\nu$, i.e.

$$(f \circ a)\, a^*\Omega = f\Omega \implies -(f \circ a)\Omega = f\Omega \implies f(-x) = -f(x),$$

contradicting constant sign. So $\mathbb{RP}^n$ is not orientable. □

8. The irrational line on the torus

Write $\gamma = p \circ \ell$ with $\ell(t) = (t, \alpha t)$ and $p : \mathbb{R}^2 \to T^2$, $p(s,u) = (e^{2\pi i s}, e^{2\pi i u})$, a smooth covering map and local diffeomorphism. $d\ell$ is injective, so $d\gamma = dp \circ d\ell$ is injective: γ is an immersion.

Injective. If $\gamma(s) = \gamma(t)$ then $s - t \in \mathbb{Z}$ and $\alpha(s-t) \in \mathbb{Z}$. If $s \neq t$, $\alpha = \alpha(s-t)/(s-t) \in \mathbb{Q}$, contradiction.

Not an embedding. The values $n\alpha \bmod 1$, $n \in \mathbb{Z}$, are pairwise distinct (by irrationality) and dense, so every interval $(0, \varepsilon)$ contains infinitely many of them. Choose distinct integers n_k with $n_k\alpha \bmod 1 \in (0, 1/k)$; since they are distinct, $|n_k| \to \infty$ along a subsequence. Then

$$\gamma(n_k) = (1, e^{2\pi i n_k \alpha}) \to (1,1) = \gamma(0).$$

If γ were a homeomorphism onto its image, we would get $n_k \to 0$, which is false. □

9. Flows

(a) Integral curves solve $\dot{x} = x^2$, $x(0) = x_0$. For $x_0 = 0$, $x \equiv 0$. Otherwise separate variables: $-1/x = t - 1/x_0$, so

$$\theta_t(x_0) = \frac{x_0}{1 - t x_0}.$$

(Check: $\frac{d}{dt}\frac{x_0}{1-tx_0} = \frac{x_0^2}{(1-tx_0)^2} = \theta_t(x_0)^2$.) The maximal interval is where $1 - tx_0 > 0$ (the solution starts there and blows up as $tx_0 \to 1$): $t < 1/x_0$ for $x_0 > 0$, $t > 1/x_0$ for $x_0 < 0$, all t for $x_0 = 0$. So

$$\mathcal{D} = \{(t, x) : tx < 1\}.$$

Since $\mathcal{D} \neq \mathbb{R} \times \mathbb{R}$ (e.g. the curve through $x_0 = 1$ exists only for $t < 1$), X is not complete.

(b) $\dot{x} = x$, $\dot{y} = 2y$ gives $\theta_t(x,y) = (e^t x, e^{2t} y)$, defined for all t. Then $\theta_s(\theta_t(x,y)) = (e^s e^t x, e^{2s} e^{2t} y) = \theta_{s+t}(x,y)$. □

10. Lie derivative of $x\,dy$ along the rotation field

The flow of $X = x\partial_y - y\partial_x$ is rotation: $\theta_t(x,y) = (x\cos t - y\sin t,\ x\sin t + y\cos t)$ (check: $\frac{d}{dt}$ at $t=0$ is $(-y,x)$).

(a)
$$\theta_t^*(x\,dy) = (x\cos t - y\sin t)\,(\sin t\,dx + \cos t\,dy).$$
Differentiating at $t=0$: $(-y)(dy) + x(dx)$. So $\mathcal{L}_X\omega = x\,dx - y\,dy$.

(b) $i_X\omega = x\cdot dy(X) = x\cdot x = x^2$, so $d(i_X\omega) = 2x\,dx$. And $d\omega = dx\wedge dy$, with $i_X(dx\wedge dy) = dx(X)\,dy - dy(X)\,dx = -y\,dy - x\,dx$. Cartan's formula:
$$\mathcal{L}_X\omega = i_X d\omega + d\,i_X\omega = (-x\,dx - y\,dy) + 2x\,dx = x\,dx - y\,dy.$$
The two computations agree. □

11. $\int_{S^2}\omega = 4\pi$

Directly. F restricted to $(0,\pi)\times(0,2\pi)$ is a diffeomorphism onto S^2 minus a closed set of measure zero (a half great circle), which doesn't affect the integral. It is orientation-preserving for the Stokes orientation: the outward normal at F is F itself and $\det(F, F_\varphi, F_\theta) = F\cdot(F_\varphi\times F_\theta) = \sin\varphi > 0$. So by Problem 3(c),
$$\int_{S^2}\omega = \int_0^{2\pi}\int_0^{\pi}\sin\varphi\,d\varphi\,d\theta = 2\pi\cdot 2 = 4\pi.$$

By Stokes. $\int_{S^2}\omega = \int_{\overline{B}^3} d\omega = 3\mathrm{vol}(B^3) = 3\cdot\frac{4}{3}\pi = 4\pi$. □

12. $H^1_{dR}(S^1)\cong\mathbb{R}$

Every 1-form on the 1-manifold S^1 is closed. Define $I:\Omega^1(S^1)\to\mathbb{R}$, $I(\omega) = \int_{S^1}\omega$ (counterclockwise orientation). I is linear and kills exact forms (Stokes, or the fundamental theorem for line integrals), so it descends to $\bar{I}: H^1_{dR}(S^1)\to\mathbb{R}$.

Surjective: the restriction of the angle form of Problem 5 has integral 2π.

Injective: let $\int_{S^1}\omega = 0$ and $p:\mathbb{R}\to S^1$, $p(t) = (\cos t,\sin t)$. Then $p^*\omega = g(t)\,dt$ with g smooth and 2π-periodic, and $\int_0^{2\pi} g = \int_{S^1}\omega = 0$. Let $F(t) = \int_0^t g$. Then $F(t+2\pi) - F(t) = \int_t^{t+2\pi} g = \int_0^{2\pi} g = 0$, so $F = f\circ p$ for a function f on S^1. f is smooth because p is a local diffeomorphism and F is smooth. Finally $p^*(df) = dF = g\,dt = p^*\omega$, and since p is a local diffeomorphism, $df = \omega$. So $[\omega] = 0$. □

13. Top-degree forms on closed manifolds

(a) Stokes' theorem with $\partial M = \emptyset$: $\int_M d\eta = \int_{\partial M}\eta = 0$ (forms on a compact manifold are compactly supported).

(b) M is oriented, so it has a positively oriented volume form μ, and $\int_M\mu > 0$ (it is positive in every oriented chart, and we assume $M\neq\emptyset$). By (a), μ is not exact. It is closed ($d\mu$ is an $(n+1)$-form on an n-manifold), so $[\mu]\neq 0$ in $H^n_{dR}(M)$.

(c) $H^n_{dR}(S^n)\neq 0$ by (b), while $H^n_{dR}(\mathbb{R}^n) = 0$ by the Poincaré lemma ($n\geq 1$). de Rham cohomology is a diffeomorphism invariant, so $S^n\not\cong\mathbb{R}^n$. The simpler reason: S^n is compact and $\mathbb{R}^n$ is not, so they aren't even homeomorphic. □

14. No submersions from compact manifolds to $\mathbb{R}^k$

Suppose $F : M \to \mathbb{R}^k$ is a smooth submersion. Submersions are open maps (by the rank theorem, F is locally a linear projection, which is open). So $F(M)$ is open in $\mathbb{R}^k$. It is also compact, hence closed, and nonempty. $\mathbb{R}^k$ is connected, so $F(M) = \mathbb{R}^k$, which is not compact ($k \geq 1$). Contradiction. □

15. Existence of Riemannian metrics

Cover M by charts $(U_\alpha, \varphi_\alpha)$ and let $g_\alpha = \varphi_\alpha^* \bar{g}$, the pullback of the Euclidean metric $\bar{g} = \sum (dx^i)^2$: a Riemannian metric on U_α (positive definite since $d\varphi_\alpha$ is injective). Let $\{\psi_\alpha\}$ be a smooth partition of unity subordinate to $\{U_\alpha\}$. Then $\psi_\alpha g_\alpha$, extended by zero outside supp ψ_α, is a smooth symmetric 2-tensor field on M, and

$$g = \sum_\alpha \psi_\alpha g_\alpha$$

is a locally finite sum, hence smooth and symmetric. For $p \in M$ and $0 \neq v \in T_pM$, every term $\psi_\alpha(p) g_\alpha(v, v)$ is ≥ 0, and some $\psi_\alpha(p) > 0$ (they sum to 1), with $p \in U_\alpha$ so $g_\alpha(v, v) > 0$. So $g_p(v, v) > 0$. □

(The same argument fails for Lorentzian metrics: convex combinations of indefinite forms can be degenerate.)

16. Parallelizability

(a) $p : \mathbb{R}^2 \to T^2$, $p(s, u) = (e^{2\pi i s}, e^{2\pi i u})$, is a local diffeomorphism and $p(x + m) = p(x)$ for $m \in \mathbb{Z}^2$. Since translations preserve ∂_s, ∂_u, the vector fields $E_1 = dp(\partial_s)$, $E_2 = dp(\partial_u)$ are well defined on T^2, smooth, and linearly independent at every point. (Alternatively: T^2 is a Lie group, and every Lie group is parallelizable by left-invariant fields.)

(b) Use $\langle p, q \rangle = \mathrm{Re}(p\bar{q})$, the Euclidean inner product on $\mathbb{H} = \mathbb{R}^4$, and $|pq| = |p||q|$. For a unit quaternion p and $u \in \{i, j, k\}$:

- $|up| = 1$;
- $\langle up, p \rangle = \mathrm{Re}(up\bar{p}) = \mathrm{Re}(u) = 0$, so $up \in p^\perp = T_pS^3$;
- $\langle ip, jp \rangle = \mathrm{Re}(ip\bar{p}\bar{j}) = \mathrm{Re}(i \cdot (-j)) = \mathrm{Re}(-k) = 0$; similarly $\langle ip, kp \rangle = \mathrm{Re}(-ik) = \mathrm{Re}(j) = 0$ and $\langle jp, kp \rangle = \mathrm{Re}(-jk) = \mathrm{Re}(-i) = 0$.

The maps $p \mapsto up$ are restrictions of linear maps, hence smooth. So (ip, jp, kp) is a global orthonormal frame on S^3.

(c) A frame on S^2 would include a nowhere-vanishing vector field, which doesn't exist by the hairy ball theorem (3.4 notes, §4.3). The quaternion trick works on S^3 (and the analogous complex trick on S^1) because these spheres are groups of unit elements in a normed division algebra; S^2 is not a Lie group, since every Lie group is parallelizable. □

Qualifying-exam level

17. $H^*_{dR}(S^n)$

Let $U = S^n \setminus \{N\}$, $V = S^n \setminus \{S\}$. Both are diffeomorphic to $\mathbb{R}^n$, so by the Poincaré lemma $H^0 = \mathbb{R}$ and $H^k = 0$ for $k \geq 1$. $U \cap V \cong S^{n-1} \times \mathbb{R}$ is homotopy equivalent to S^{n-1}, so by homotopy invariance $H^k(U \cap V) \cong H^k(S^{n-1})$.

$n = 1$. $U \cap V$ is two disjoint intervals: $H^0(U \cap V) = \mathbb{R}^2$, $H^1 = 0$. The Mayer–Vietoris sequence reads

$$0 \to H^0(S^1) \to H^0(U) \oplus H^0(V) \xrightarrow{\beta} H^0(U \cap V) \xrightarrow{\delta} H^1(S^1) \to H^1(U) \oplus H^1(V) = 0.$$

$H^0(S^1) = \mathbb{R}$ (connected). Dimensions: $\mathbb{R} \to \mathbb{R}^2 \to \mathbb{R}^2 \to H^1(S^1) \to 0$. Exactness gives $\dim \operatorname{im} \beta = 2 - 1 = 1$, so $\dim H^1(S^1) = 2 - 1 = 1$. And $H^k(S^1) = 0$ for $k \geq 2$ by dimension.

$n \geq 2$. All of $S^n, U, V, U \cap V$ are connected.

- Degree 0–1: $0 \to \mathbb{R} \to \mathbb{R}^2 \xrightarrow{\beta} \mathbb{R} \xrightarrow{\delta} H^1(S^n) \to 0$. $\beta(a, b) = a - b$ is surjective, so $\delta = 0$ and $H^1(S^n) = 0$.
- For $k \geq 2$: $0 = H^{k-1}(U) \oplus H^{k-1}(V) \to H^{k-1}(U \cap V) \xrightarrow{\delta} H^k(S^n) \to H^k(U) \oplus H^k(V) = 0$, so $H^k(S^n) \cong H^{k-1}(S^{n-1})$.

By induction from $n = 1$: $H^k_{dR}(S^n) = \mathbb{R}$ for $k = 0$ and $k = n$, and 0 otherwise. □

18. Smooth Brouwer

(a) Suppose $r : \overline{B}^n \to S^{n-1}$ is smooth with $r \circ \iota = \mathrm{id}$, where $\iota : S^{n-1} = \partial \overline{B}^n \hookrightarrow \overline{B}^n$. Let σ be a volume form on S^{n-1} positive for the Stokes orientation ($n - 1 \geq 1$), so $\int_{S^{n-1}} \sigma > 0$. Then

$$0 < \int_{S^{n-1}} \sigma = \int_{S^{n-1}} \iota^* r^* \sigma = \int_{\overline{B}^n} d(r^* \sigma) = \int_{\overline{B}^n} r^*(d\sigma) = 0,$$

since $d\sigma$ is an n-form on the $(n-1)$-manifold S^{n-1}, hence zero. Contradiction.

(b) Let $f : \overline{B}^n \to \overline{B}^n$ be smooth with $f(x) \neq x$ for all x. Put $u(x) = x - f(x) \neq 0$ and look for $t(x) \geq 0$ with $|x + t\,u(x)| = 1$:

$$t(x) = \frac{-(x \cdot u) + \sqrt{(x \cdot u)^2 + |u|^2(1 - |x|^2)}}{|u|^2}.$$

The discriminant is positive on the open ball (second term > 0). On the sphere $|x| = 1$ it equals $(x \cdot u)^2$, and $x \cdot u = 1 - x \cdot f(x) \geq 1 - |f(x)| \geq 0$, with equality only if $x \cdot f(x) = 1$, which (Cauchy–Schwarz, $|x| = 1$, $|f(x)| \leq 1$) forces $f(x) = x$. So $x \cdot u > 0$ on the sphere, the discriminant is positive everywhere, and t is smooth. On the sphere $t(x) = \frac{-(x \cdot u) + |x \cdot u|}{|u|^2} = 0$. So $r(x) = x + t(x)u(x)$ is a smooth retraction $\overline{B}^n \to S^{n-1}$, contradicting (a). □

(The continuous Brouwer theorem then follows by approximating a continuous map uniformly by smooth ones, e.g. via Stone–Weierstrass, and using compactness.)

19. Closed 1-forms on simply connected manifolds are exact

Let ω be closed. By the Poincaré lemma, every point has a coordinate ball B on which $\omega = df_B$ for some smooth f_B.

Step 1 (homotopy invariance of line integrals). Let $\gamma_0, \gamma_1 : [0, 1] \to M$ be piecewise smooth paths from p to q, and $H : I \times I \to M$ a continuous path homotopy ($H(\cdot, 0) = \gamma_0$, $H(\cdot, 1) = \gamma_1$, $H(0, s) = p$, $H(1, s) = q$). By the Lebesgue number lemma, subdivide $I \times I$ into small rectangles R, each mapped by H into some ball B_R with primitive f_R. For an edge e of the grid from vertex v to vertex w lying in a rectangle R, set $c(e) = f_R(H(w)) - f_R(H(v))$. If e lies in two rectangles R, R', then $f_R - f_{R'}$ has zero differential on $B_R \cap B_{R'}$, so it is constant on the connected set

$H(e)$; hence $c(e)$ is independent of the choice of R. Summing c around the boundary of a single rectangle gives 0 (it telescopes with the single function f_R). Summing over all rectangles, interior edges cancel, so the sum of c around $\partial(I \times I)$ is 0. The vertical sides contribute 0 (they map to the points p, q). On the bottom edge, each piece $\gamma_0|_{[t_j,t_{j+1}]}$ lies in B_R, where $\int \omega = \int df_R = c(e)$; so the bottom contributes $\int_{\gamma_0} \omega$, and likewise the top contributes $-\int_{\gamma_1} \omega$. Hence $\int_{\gamma_0} \omega = \int_{\gamma_1} \omega$.

Step 2 (define the primitive). Fix p_0. A connected manifold is connected by piecewise smooth paths (the set of points so reachable from p_0 is open and closed, using coordinate balls). Define $f(p) = \int_\gamma \omega$ for any piecewise smooth γ from p_0 to p. Since M is simply connected, any two such paths are path-homotopic, so by Step 1 f is well defined.

Step 3 ($df = \omega$). Fix p and a ball $B \ni p$ with $\omega = dg$ on B. For $q \in B$, concatenate a path to p with a path inside B from p to q: $f(q) = f(p) + g(q) - g(p)$. So $f = g +$ const on B, hence f is smooth and $df = dg = \omega$ there. □

20. A non-integrable distribution

(a) $\alpha = dz - y\,dx$ satisfies $\alpha(V) = y - y = 0$ and $\alpha(W) = 0$. α is nowhere zero, so $\ker\alpha$ has rank 2, and V, W are everywhere linearly independent and lie in it. So $D = \ker\alpha$.

(b) For any f,

$$[V,W]f = (\partial_x + y\partial_z)(f_y) - \partial_y(f_x + yf_z) = f_{xy} + yf_{zy} - f_{xy} - f_z - yf_{zy} = -f_z,$$

so $[V,W] = -\partial_z$. Since $\alpha(-\partial_z) = -1 \neq 0$, $[V,W] \notin D$ at every point. D is not involutive. An integrable distribution is involutive (if S is an integral surface, V, W are tangent to S, so their bracket is tangent to S), so D is not integrable.

(c) $d\alpha = -dy \wedge dx = dx \wedge dy$ and $\alpha \wedge d\alpha = dz \wedge dx \wedge dy = dx \wedge dy \wedge dz \neq 0$.

Why it detects (b). For sections V, W of $\ker\alpha$, $d\alpha(V,W) = V(\alpha(W)) - W(\alpha(V)) - \alpha([V,W]) = -\alpha([V,W])$. So $\ker\alpha$ is involutive iff $d\alpha$ vanishes on $\ker\alpha$. Choosing T with $\alpha(T) = 1$, $(\alpha \wedge d\alpha)(T,V,W) = \alpha(T)\,d\alpha(V,W) = d\alpha(V,W)$ (the other terms contain $\alpha(V)$ or $\alpha(W)$). So $\ker\alpha$ is involutive iff $\alpha \wedge d\alpha = 0$. Here $d\alpha(V,W) = 1$, matching $-\alpha([V,W]) = -\alpha(-\partial_z) = 1$. α is the standard **contact form** on $\mathbb{R}^3$. □

21. Commuting flows

($\Rightarrow$) Assume $[X,Y] = 0$. Fix p and set $V(t) = d(\theta_{-t})_{\theta_t(p)}(Y_{\theta_t(p)}) \in T_pM$. By the given formula, $V'(t_0) = d(\theta_{-t_0})([X,Y]_{\theta_{t_0}(p)}) = 0$ for all t_0, so $V(t) = V(0) = Y_p$. Applying $d\theta_t$: $d(\theta_t)_p(Y_p) = Y_{\theta_t(p)}$ for all p, t, i.e. Y is θ_t-invariant.

Now fix t and q, and let $c(s) = \theta_t(\psi_s(q))$. Then

$$c'(s) = d\theta_t(Y_{\psi_s(q)}) = Y_{\theta_t(\psi_s(q))} = Y_{c(s)},$$

so c is an integral curve of Y with $c(0) = \theta_t(q)$. By uniqueness of integral curves, $c(s) = \psi_s(\theta_t(q))$. So $\theta_t \circ \psi_s = \psi_s \circ \theta_t$.

($\Leftarrow$) If the flows commute, differentiate $\theta_t(\psi_s(q)) = \psi_s(\theta_t(q))$ in s at $s = 0$: $d\theta_t(Y_q) = Y_{\theta_t(q)}$ for all t, q. Then $V(t) = d(\theta_{-t})(Y_{\theta_t(p)}) = Y_p$ is constant, so $[X,Y]_p = \mathcal{L}_X Y(p) = V'(0) = 0$. □

22. $SL(n,\mathbb{R})$ and $\det\exp = \exp\operatorname{tr}$

(a) $\det(A + tB) = \det(A)\det(I + tA^{-1}B)$. For any matrix C, $\det(I + tC) = 1 + t\operatorname{tr}C + O(t^2)$ (expand the determinant: the only terms of order ≤ 1 in t come from the product of diagonal

entries). So

$$d(\det)_A(B) = \frac{d}{dt}\Big|_0 \det(A+tB) = \det(A)\mathrm{tr}(A^{-1}B).$$

(b) For $A \in SL(n)$ and $c \in \mathbb{R}$, $d(\det)_A(cA/n) = \mathrm{tr}(cI/n) = c$. So $d(\det)_A$ is onto $\mathbb{R}$ and 1 is a regular value of $\det : GL(n) \to \mathbb{R} \setminus 0$. Therefore $SL(n) = \det^{-1}(1)$ is an embedded submanifold of dimension $n^2 - 1$, and $T_I SL(n) = \ker d(\det)_I = \{B : \mathrm{tr}\, B = 0\}$.

(c) Let $g(t) = \det(e^{tB})$. Since $\frac{d}{dt}e^{tB} = Be^{tB}$, the chain rule and (a) give

$$g'(t) = \det(e^{tB})\mathrm{tr}\left(e^{-tB}Be^{tB}\right) = g(t)\mathrm{tr}\, B,$$

using $\mathrm{tr}(P^{-1}BP) = \mathrm{tr}\, B$. With $g(0) = 1$, $g(t) = e^{t\,\mathrm{tr}\, B}$; at $t = 1$, $\det e^B = e^{\mathrm{tr}\, B}$. In particular $\mathrm{tr}\, B = 0$ implies $\det e^B = 1$, so exp maps $\mathfrak{sl}(n, \mathbb{R})$ into $SL(n, \mathbb{R})$. □